



APPLIED



NUMERICAL



LINEAR



ALGEBRA

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siam

3.2.3. Singular Value Decomposition

The SVD is a very important decomposition which is used for many purposes other than solving least squares problems.

THEOREM 3.2. SVD. *Let A be an arbitrary m -by- n matrix with $m \geq n$. Then we can write $A = U\Sigma V^T$, where U is m -by- n and satisfies $U^T U = I$, V is n -by- n and satisfies $V^T V = I$, and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$, where $\sigma_1 \geq \dots \geq \sigma_n \geq 0$. The columns $u_1, \dots, u_n$ of U are called left singular vectors. The columns $v_1, \dots, v_n$ of V are called right singular vectors. The σ_i are called singular values. (If $m < n$, the SVD is defined by considering A^T .)*

A geometric restatement of this theorem is as follows. Given any m -by- n matrix A , think of it as mapping a vector $x \in \mathbb{R}^n$ to a vector $y = Ax \in \mathbb{R}^m$. Then we can choose one orthogonal coordinate system for $\mathbb{R}^n$ (where the unit axes are the columns of V) and another orthogonal coordinate system for $\mathbb{R}^m$ (where the unit axes are the columns of U) such that A is diagonal (Σ), i.e., maps a vector $x = \sum_{i=1}^n \beta_i v_i$ to $y = Ax = \sum_{i=1}^n \sigma_i \beta_i u_i$. In other words, any matrix is diagonal, provided that we pick appropriate orthogonal coordinate systems for its domain and range.

Proof of Theorem 3.2. We use induction on m and n : we assume that the SVD exists for $(m-1)$ -by- $(n-1)$ matrices and prove it for m -by- n . We assume $A \neq 0$; otherwise we can take $\Sigma = 0$ and let U and V be arbitrary orthogonal matrices.

The basic step occurs when $n = 1$ (since $m \geq n$). We write $A = U\Sigma V^T$ with $U = A/\|A\|_2$, $\Sigma = \|A\|_2$, and $V = 1$.

For the induction step, choose v so $\|v\|_2 = 1$ and $\|Av\|_2 = \|A\|_2 > 0$. Such a v exists by the definition of $\|A\|_2 = \max_{\|v\|_2=1} \|Av\|_2$. Let $u = \frac{Av}{\|Av\|_2}$, which is a unit vector. Choose $\tilde{U}$ and $\tilde{V}$ so that $U = [u, \tilde{U}]$ is an m -by- m orthogonal matrix, and $V = [v, \tilde{V}]$ is an n -by- n orthogonal matrix. Now write

$$U^T AV = \begin{bmatrix} u^T \\ \tilde{U}^T \end{bmatrix} \cdot A \cdot \begin{bmatrix} v & \tilde{V} \end{bmatrix} = \begin{bmatrix} u^T Av & u^T A\tilde{V} \\ \tilde{U}^T Av & \tilde{U}^T A\tilde{V} \end{bmatrix}.$$

Then

$$u^T Av = \frac{(Av)^T(Av)}{\|Av\|_2} = \frac{\|Av\|_2^2}{\|Av\|_2} = \|Av\|_2 = \|A\|_2 \equiv \sigma$$

and $\tilde{U}^T Av = \tilde{U}^T u \|Av\|_2 = 0$. We claim $u^T A\tilde{V} = 0$ too because otherwise $\sigma = \|A\|_2 = \|U^T AV\|_2 \geq \|[1, 0, \dots, 0]U^T AV\|_2 = \|[\sigma | u^T A\tilde{V}]\|_2 > \sigma$, a contradiction. (We have used part 7 of Lemma 1.7.)

So $U^T AV = \begin{bmatrix} \sigma & 0 \\ 0 & \tilde{U}^T A\tilde{V} \end{bmatrix} = \begin{bmatrix} \sigma & 0 \\ 0 & \tilde{A} \end{bmatrix}$. We may now apply the induction hypothesis to $\tilde{A}$ to get $\tilde{A} = U_1 \Sigma_1 V_1^T$, where U_1 is $(m-1)$ -by- $(n-1)$, Σ_1 is $(n-1)$ -by- $(n-1)$, and V_1 is $(n-1)$ -by- $(n-1)$. So

$$U^T AV = \begin{bmatrix} \sigma & 0 \\ 0 & U_1 \Sigma_1 V_1^T \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & U_1 \end{bmatrix} \begin{bmatrix} \sigma & 0 \\ 0 & \Sigma_1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & V_1 \end{bmatrix}^T$$

or

$$A = \left(U \begin{bmatrix} 1 & 0 \\ 0 & U_1 \end{bmatrix} \right) \begin{bmatrix} \sigma & 0 \\ 0 & \Sigma_1 \end{bmatrix} \left(V \begin{bmatrix} 1 & 0 \\ 0 & V_1 \end{bmatrix} \right)^T,$$

which is our desired decomposition. $\square$

The SVD has a large number of important algebraic and geometric properties, the most important of which we state here.

THEOREM 3.3. Let $A = U\Sigma V^T$ be the SVD of the m -by- n matrix A , where $m \geq n$. (There are analogous results for $m < n$.)

1. Suppose that A is symmetric, with eigenvalues λ_i and orthonormal eigenvectors u_i . In other words $A = U\Lambda U^T$ is an eigendecomposition of A , with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, $U = [u_1, \dots, u_n]$, and $UU^T = I$. Then an SVD of A is $A = U\Sigma V^T$, where $\sigma_i = |\lambda_i|$ and $v_i = \text{sign}(\lambda_i)u_i$, where $\text{sign}(0) = 1$.

2. The eigenvalues of the symmetric matrix $A^T A$ are σ_i^2 . The right singular vectors v_i are corresponding orthonormal eigenvectors.
3. The eigenvalues of the symmetric matrix AA^T are σ_i^2 and $m-n$ zeroes. The left singular vectors u_i are corresponding orthonormal eigenvectors for the eigenvalues σ_i^2 . One can take any $m-n$ other orthogonal vectors as eigenvectors for the eigenvalue 0.
4. Let $H = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$, where A is square and $A = U\Sigma V^T$ is the SVD of A . Let $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$, $U = [u_1, \dots, u_n]$, and $V = [v_1, \dots, v_n]$. Then the $2n$ eigenvalues of H are $\pm\sigma_i$, with corresponding unit eigenvectors $\frac{1}{\sqrt{2}} \begin{bmatrix} v_i \\ \pm u_i \end{bmatrix}$.
5. If A has full rank, the solution of $\min_x \|Ax - b\|_2$ is $x = V\Sigma^{-1}U^T b$.
6. $\|A\|_2 = \sigma_1$. If A is square and nonsingular, then $\|A^{-1}\|_2^{-1} = \sigma_n$ and $\|A\|_2 \cdot \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}$.
7. Suppose $\sigma_1 \geq \dots \geq \sigma_r > \sigma_{r+1} = \dots = \sigma_n = 0$. Then the rank of A is r . The null space of A , i.e., the subspace of vectors v such that $Av = 0$, is the space spanned by columns $r+1$ through n of V : $\text{span}(v_{r+1}, \dots, v_n)$. The range space of A , the subspace of vectors of the form Aw for all w , is the space spanned by columns 1 through r of U : $\text{span}(u_1, \dots, u_r)$.
8. Let S^{n-1} be the unit sphere in $\mathbb{R}^n$: $S^{n-1} = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$. Let $A \cdot S^{n-1}$ be the image of S^{n-1} under A : $A \cdot S^{n-1} = \{Ax : x \in \mathbb{R}^n \text{ and } \|x\|_2 = 1\}$. Then $A \cdot S^{n-1}$ is an ellipsoid centered at the origin of $\mathbb{R}^m$, with principal axes $\sigma_i u_i$.
9. Write $V = [v_1, v_2, \dots, v_n]$ and $U = [u_1, u_2, \dots, u_m]$, so $A = U\Sigma V^T = \sum_{i=1}^n \sigma_i u_i v_i^T$ (a sum of rank-1 matrices). Then a matrix of rank $k < n$ closest to A (measured with $\|\cdot\|_2$) is $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$, and $\|A - A_k\|_2 = \sigma_{k+1}$. We may also write $A_k = U\Sigma_k V^T$, where $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0)$.

Proof.

1. This is true by the definition of the SVD.
2. $A^T A = V\Sigma U^T U\Sigma V^T = V\Sigma^2 V^T$. This is an eigendecomposition of $A^T A$, with the columns of V the eigenvectors and the diagonal entries of Σ^2 the eigenvalues.
3. Choose an m -by- $(m-n)$ matrix $\tilde{U}$ so that $[U, \tilde{U}]$ is square and orthogonal. Then write

$$AA^T = U\Sigma V^T V\Sigma U^T = U\Sigma^2 U^T = [U, \tilde{U}] \begin{bmatrix} \Sigma^2 & 0 \\ 0 & 0 \end{bmatrix} [U, \tilde{U}]^T.$$

This is an eigendecomposition of AA^T .

4. See Question 3.14.

5. $\|Ax - b\|_2^2 = \|U\Sigma V^T x - b\|_2^2$. Since A has full rank, so does Σ , and thus Σ is invertible. Now let $[U, \tilde{U}]$ be square and orthogonal as above so

$$\begin{aligned} \|U\Sigma V^T x - b\|_2^2 &= \left\| \begin{bmatrix} U^T \\ \tilde{U}^T \end{bmatrix} (U\Sigma V^T x - b) \right\|_2^2 \\ &= \left\| \begin{bmatrix} \Sigma V^T x - U^T b \\ -\tilde{U}^T b \end{bmatrix} \right\|_2^2 \\ &= \|\Sigma V^T x - U^T b\|_2^2 + \|\tilde{U}^T b\|_2^2. \end{aligned}$$

This is minimized by making the first term zero, i.e., $x = V\Sigma^{-1}U^T b$.

6. It is clear from its definition that the two-norm of a diagonal matrix is the largest absolute entry on its diagonal. Thus, by part 3 of Lemma 1.7, $\|A\|_2 = \|U^T A V\|_2 = \|\Sigma\|_2 = \sigma_1$ and $\|A^{-1}\|_2 = \|V^T A^{-1} U\|_2 = \|\Sigma^{-1}\|_2 = \sigma_n^{-1}$.

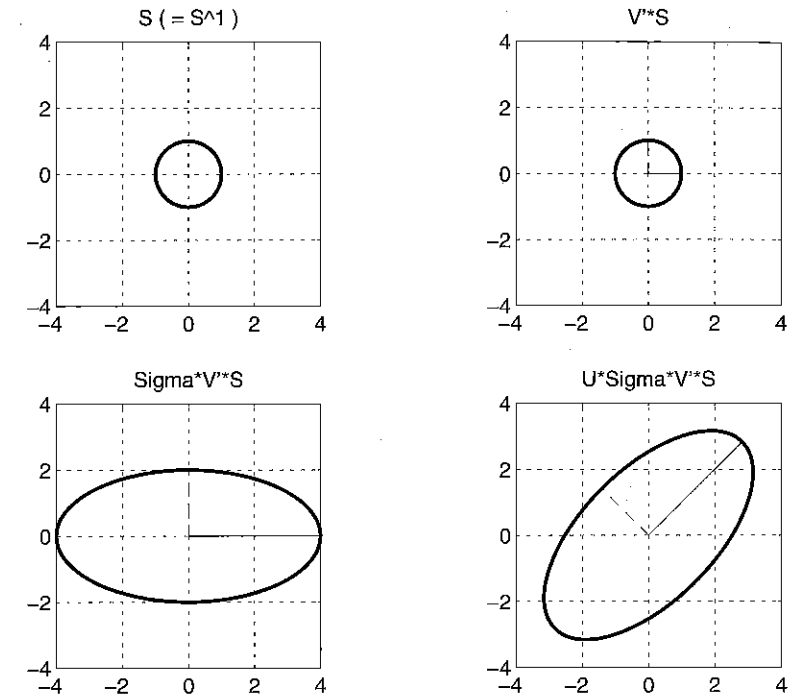
7. Again choose an m -by- $(m-n)$ matrix $\tilde{U}$ so that the m -by- m matrix $\hat{U} = [U, \tilde{U}]$ is orthogonal. Since $\hat{U}$ and V are nonsingular, A and $\hat{U}^T A V = \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} \equiv \hat{\Sigma}$ have the same rank—namely, r —by our assumption about Σ . Also, v is in the null space of A if and only if $V^T v$ is in the null space of $\hat{U}^T A V = \hat{\Sigma}$, since $Av = 0$ if and only if $\hat{U}^T A V(V^T v) = 0$. But the null space of $\hat{\Sigma}$ is clearly spanned by columns $r+1$ through n of the n -by- n identity matrix I_n , so the null space of A is spanned by V times these columns, i.e., v_{r+1} through v_n . A similar argument shows that the range space of A is the same as $\hat{U}$ times the range space of $\hat{U}^T A V = \hat{\Sigma}$, i.e., $\hat{U}$ times the first r columns of I_m , or u_1 through u_r .

8. We “build” the set $A \cdot S^{n-1}$ by multiplying by one factor of $A = U\Sigma V^T$ at a time. The figure below illustrates what happens when

$$\begin{aligned} A &= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2^{-1/2} & -2^{-1/2} \\ 2^{-1/2} & 2^{-1/2} \end{bmatrix} \cdot \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2^{-1/2} & -2^{-1/2} \\ 2^{-1/2} & 2^{-1/2} \end{bmatrix}^T \\ &\equiv U\Sigma V^T. \end{aligned}$$

Assume for simplicity that A is square and nonsingular. Since V is orthogonal and so maps unit vectors to other unit vectors, $V^T \cdot S^{n-1} = S^{n-1}$. Next, since $v \in S^{n-1}$ if and only if $\|v\|_2 = 1$, $w \in \Sigma S^{n-1}$ if and only if $\|\Sigma^{-1}w\|_2 = 1$ or $\sum_{i=1}^n (w_i/\sigma_i)^2 = 1$. This defines an ellipsoid with

principal axes $\sigma_i e_i$, where e_i is the i th column of the identity matrix. Finally, multiplying each $w = \Sigma v$ by U just rotates the ellipse so that each e_i becomes u_i , the i th column of U .



9. A_k has rank k by construction and

$$\|A - A_k\|_2 = \left\| \sum_{i=k+1}^n \sigma_i u_i v_i^T \right\| = \left\| U \begin{bmatrix} 0 & & \\ & \sigma_{k+1} & \\ & & \ddots \\ & & & \sigma_n \end{bmatrix} V^T \right\|_2 = \sigma_{k+1}.$$

It remains to show that there is no closer rank k matrix to A . Let B be any rank k matrix, so its null space has dimension $n - k$. The space spanned by $\{v_1, \dots, v_{k+1}\}$ has dimension $k + 1$. Since the sum of their dimensions is $(n - k) + (k + 1) > n$, these two spaces must overlap. Let h be a unit vector in their intersection. Then

$$\begin{aligned} \|A - B\|_2^2 &\geq \|(A - B)h\|_2^2 = \|Ah\|_2^2 = \|U\Sigma V^T h\|_2^2 \\ &= \|\Sigma(V^T h)\|_2^2 \\ &\geq \sigma_{k+1}^2 \|V^T h\|_2^2 \\ &= \sigma_{k+1}^2. \quad \square \end{aligned}$$

EXAMPLE 3.4. We illustrate the last part of Theorem 3.3 by using it for *image compression*. In particular, we will illustrate it with low-rank approximations

of a clown. An m -by- n image is just an m -by- n matrix, where entry (i, j) is interpreted as the brightness of pixel (i, j) . In other words, matrix entries ranging from 0 to 1 (say) are interpreted as pixels ranging from black (=0) through various shades of gray to white (=1). (Colors also are possible.) Rather than storing or transmitting all $m \cdot n$ matrix entries to represent the image, we often prefer to *compress* the image by storing many fewer numbers, from which we can still approximately reconstruct the original image. We may use Part 9 of Theorem 3.3 to do this, as we now illustrate.

Consider the image in Figure 3.3(a). This 320-by-200 pixel image corresponds to a 320-by-200 matrix A . Let $A = U\Sigma V^T$ be the SVD of A . Part 9 of Theorem 3.3 tells us that $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$ is the best rank- k approximation of A , in the sense of minimizing $\|A - A_k\|_2 = \sigma_{k+1}$. Note that it only takes $m \cdot k + n \cdot k = (m + n) \cdot k$ words to store u_1 through u_k and $\sigma_1 v_1$ through $\sigma_k v_k$, from which we can reconstruct A_k . In contrast, it takes $m \cdot n$ words to store A (or A_k explicitly), which is much larger when k is small. So we will use A_k as our compressed image, stored using $(m + n) \cdot k$ words. The other images in Figure 3.3 show these approximations for various values of k , along with the relative errors σ_{k+1}/σ_1 and compression ratios $(m + n) \cdot k / (m \cdot n) = 520 \cdot k / 64000 \approx k/123$.

k	Relative error = σ_{k+1}/σ_k	Compression ratio = $520k/64000$
3	.155	.024
10	.077	.081
20	.040	.163

These images were produced by the following commands (the clown and other images are available in Matlab among the visualization demonstration files; check your local installation for location):

```
load clown.mat; [U,S,V]=svd(X); colormap('gray');
image(U(:,1:k)*S(1:k,1:k)*V(:,1:k))
```

There are also many other, cheaper image-compression techniques available than the SVD [189, 152]. $\diamond$

Later we will see that the cost of solving a least squares problem with the SVD is about the same as with QR when $m \gg n$, and about $4n^2m - \frac{4}{3}n^3 + O(n^2)$ for smaller m . A precise comparison of the costs of QR and the SVD also depends on the machine being used. See section 3.6 for details.

DEFINITION 3.1. Suppose that A is m -by- n with $m \geq n$ and has full rank, with $A = QR = U\Sigma V^T$ being A 's QR decomposition and SVD, respectively. Then

$$A^+ \equiv (A^T A)^{-1} A^T = R^{-1} Q^T = V \Sigma^{-1} U^T$$

is called the (Moore-Penrose) pseudoinverse of A . If $m < n$, then $A^+ \equiv A^T (A A^T)^{-1}$.

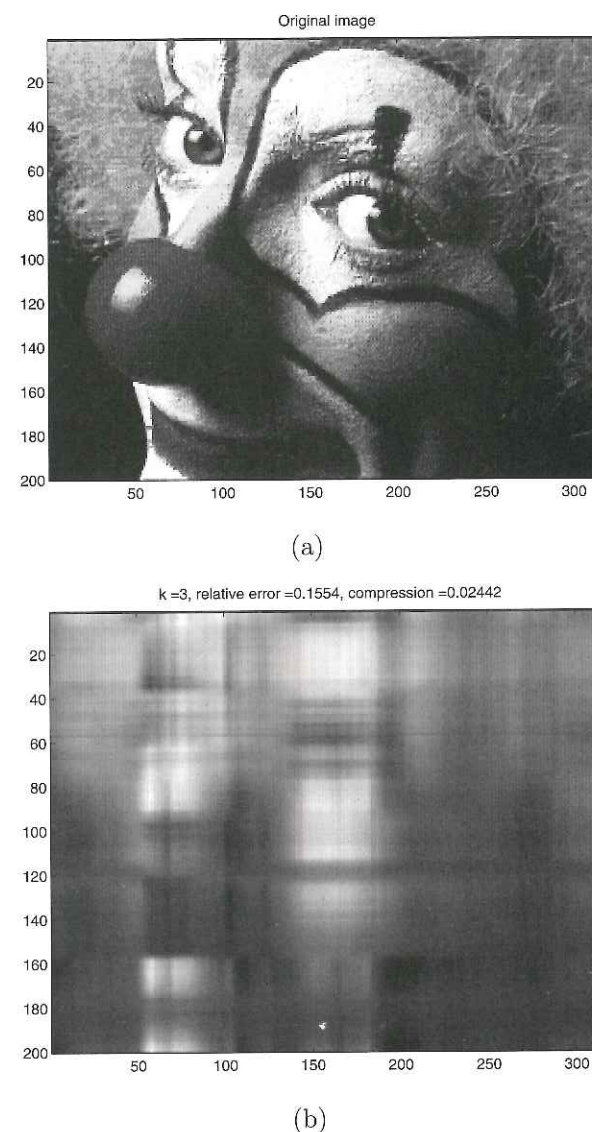
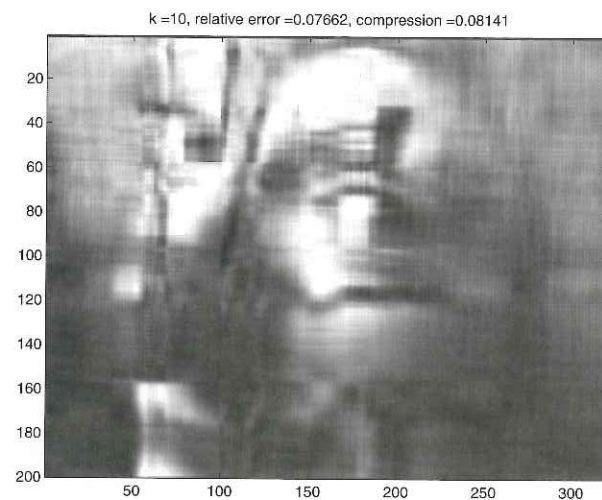
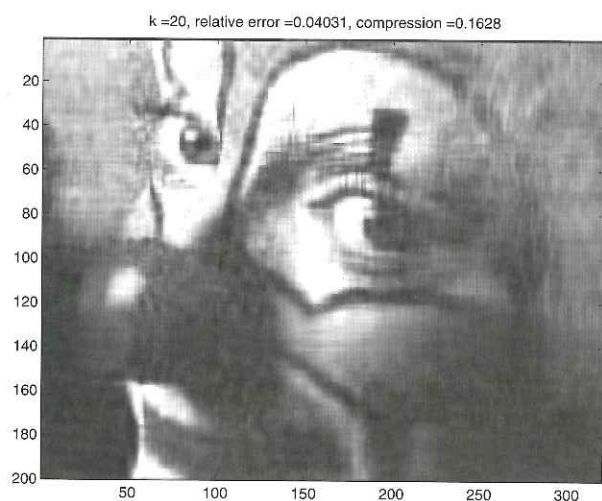


Fig. 3.3. Image compression using the SVD. (a) Original image. (b) Rank $k = 3$ approximation.



(c)



(d)

Fig. 3.3. *Continued.* (c) Rank $k = 10$ approximation. (d) Rank $k = 20$ approximation.

The pseudoinverse lets us write the solution of the full-rank, overdetermined least squares problem as simply $x = A^+b$. If A is square and full rank, this formula reduces to $x = A^{-1}b$ as expected. The pseudoinverse of A is computed as `pinv(A)` in Matlab. When A is not full rank, the Moore–Penrose pseudoinverse is given by Definition 3.2 in section 3.5.